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The growing partisan politicization of non-political online spaces: A mixed-method analysis of news app reviews on Google Play between 2009 and 2022

Rui Wang
University at Buffalo, USA

Sagarika Suresh Thimmanayakanapalya
University at Buffalo, USA

Yotam Ophir
University at Buffalo, USA

Abstract

Drawing on theories of identity politics and partisan polarization, we explored the politicization of Google Play's news app reviews—an explicitly non-political domain. Using a mixed-methods approach, Analysis of Topic Model Networks (ANTMNs), combining topic modeling, network analysis, community detection, and theory-driven qualitative reading, we analyzed 759,143 reviews from 2009 to 2022 across 46 news apps. Three themes emerged: Technical, Content Quality, and Political. The political discourse in reviews has intensified over the years, with notable spikes around election periods. Accusations of bias were found to correlate most strongly with lower app ratings. The findings provide alarming empirical evidence for the politicization of non-political spaces, such as the app reviews section on app stores. With identity politics on the rise, this study sheds light on the importance of considering non-political online spaces for the study of political discourse.

Corresponding author:
Rui Wang, Department of Communication, University at Buffalo, 359 Baldy Hall, Buffalo, NY 14260, USA.
Email: rwang38@buffalo.edu

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In recent years, scholars have expressed growing concerns about partisan divisions among Americans (Radwell and Israel, 2021), metastasized beyond lines of ideology, into domains of race, geography, and religion (Mason, 2018). As these schisms have become increasingly associated with the two main political parties (i.e. political sorting), many aspects that were once considered non-political have become politicized. From mask-wearing during coronavirus disease (COVID)-19 (Shin et al., 2022) and vaccination (Walter et al., 2020) to dating (Huber and Malhotra, 2017) and food preferences (Keefe, 2020), Americans' behaviors could now be effectively predicted from their identification as either Republicans or Democrats.

Political polarization has influenced Americans' media consumption as well. Up until the 1980s, Americans of all political affiliations have gotten most of their news from the three big networks (ABC, CBS, and NBC), several local or national newspapers, or the mainstream CNN (Napoli, 2020). With the rise of AM talk-radio, cable television networks, the Internet, and social media, it became substantially simpler for people to surround themselves with like-minded media (Jamieson and Cappella, 2008; Mitchell et al., 2021). By the 2010s, however, citizens could easily select and choose available news outlets and programs tailored to their tastes and prior political beliefs (Peterson et al., 2021). Following these developments, scholars have warned that the selection of like-minded news content and avoidance of incongruent views (Mutz, 2006) could lead to the adoption of more extreme views, which, in turn, could further increase selective exposure (Jamieson and Cappella, 2008). These processes could result in a cycle of polarization (Levendusky, 2013).

Considering these processes, we investigate how news consumers evaluate and discuss news outlets, with a particular attention to increasing trends in partisan politicization. We analyze 759,143 reviews published in reference to 46 news applications (apps) on Google Play, the biggest app store in the world, serving Android users. Notably, the app reviews section of Google Play is not designed for consumers to comment on the political nature of apps. Instead, the section is explicitly described by Google as a space for users to provide technical feedback to app developers. According to the "Ratings and Reviews on Google Play" page (Google and Play, 2023), reviews should address issues related to user experience and technical issues, for developers to consume and act upon. In fact, according to Google play, "solicitations, political or social commentary," which are not relevant to the "core of the consumer experience" should be considered off-topic. It is important to note that the Apple Store does not have such explicit guidelines regarding the content of app reviews. This difference makes the Google Play platform more aligned with our study's focus and theoretical framework, as it provides a clearer distinction between reviews that adhere to the intended purpose of the platform and those that deviate into political or social commentary.

Despite Google's intentions, we anticipated a growing politicization of news app reviews. This expectation is based on observations made by other scholars in other traditionally non-political areas such as medicine (Ophir et al., 2023) and science (Walter

et al., 2022). Given the scope of the data and the lack of prior research into similar phenomena, we approach these questions using an inductive mixed-method framework, the Analysis of Topic Model Networks (ANTMNs), combining unsupervised machine learning and in-depth manual coding (Walter and Ophir, 2019).

Identity politics, mega-identities, and the politicization of everyday life

The emotional and cognitive attachment to political parties has become omnipresent in American politics since the 1960s (Iyengar and Westwood, 2015). As argued by Mason (2018), Americans did not necessarily become more extreme in policy preferences. Instead, cultural and social preferences and traits have become significantly more "sorted" into two increasingly homogeneous party bins. Ideologically, the Republican party has become consistently more conservative, and to a much lesser degree, the Democratic more liberal (as evident, for example, in Pew Research data),[1] making it harder for politicians to reach agreements and cooperation across the aisle (Van Boven et al., 2018). Importantly, this political sorting extends beyond ideology, but rather spilled over to race, gender, age, income, education, and culture. As evident in myriad studies, Republicans and Democrats now drink different types of coffee, shop in different stores, and consume different art and music (Young, 2019). Democrats are more likely to shop at the likes of H&M, Starbucks, and Target, while Republicans prefer Walmart, Chick-fil-A, and Hobby Lobby (Lee, 2020).

Moreover, political identities are increasingly intertwined with the choice of news and social media platforms, with liberals consuming a more diverse and balanced media diet, whereas conservatives relying heavily on a few sources they trust, despite studies showing them to be biased and prone to misinformation (Benkler et al., 2018; Kor-Sins, 2023). Following decades of concerted efforts to undermine the mainstream media and other official sources of knowledge, conservatives and Republicans are now highly suspicious of mainstream media (Bedingfield, 2018; Jamieson and Cappella, 2018), and are more likely to isolate themselves in echo chambers (Levendusky et al., 2023).

Finally, sorting has made political parties a more prominent part of Americans' social identity (Turner, 2010), with politics being perceived as a zero-sum-game. The myriad consequences of sorting on anything from policy-change to lifestyle choices, what scholars have dubbed meta-identities, has been well-documented in the political science literature (Finkel et al., 2020).

Blurring the lines between political and non-political online spaces

Google's request for users to avoid politics in reviews (Google Play, 2023) begs the question of what should be considered political in the first place. As Mutz (2001) suggested, defining politics has long been a daunting task, one that is becoming increasingly harder to demarcate in today's climate of politicization and polarization. Feminist scholars, for example, have famously argued that the term "political" should not be limited to electoral issues, but needs to be expanded to broader aspects of everyday life affected by power relations (Hanisch, 1969; Suk et al., 2023). Due to our focus on partisan identities, we follow conceptualizations of politicization as the signaling or communication of

collective identities that stand in opposition to each other, expressed in discourse of us versus them (Bakardjieva, 2015), often associated with moralistic language of right versus wrong (Mouffe, 2005). When studying the bipartisan American system (Mason, 2018), we are particularly interested in the Republican/conservative versus Democrat/liberal dichotomy. Our definition is not limited to explicit expressions of electoral preferences (Riedl et al., 2023), but rather to any communication that interprets events, people, or social phenomena in ways that project one's association with or disdain toward one of the two political parties and its followers (Sunstein, 2018).

Empirical evidence has supported a concerning growth in partisan politicization and polarization of non-political online spaces. Content analyses of online non-political chat rooms and blogs, for example, had revealed the existence of a substantial number of political discussions (Munson and Resnick, 2011; Wojcieszak and Mutz, 2009). A study combining content analysis and survey data on a popular, entertainment-oriented, subreddit focused on The Bachelor television show found a large number of political discussions as well, with users often expressing intolerance toward those with opposing political views (Chen and McCabe, 2022). Another study demonstrated that nearly half of the political discussions on Reddit are coming from non-political subreddits (Rajadesingan et al., 2021). We argue that the app review section could serve as one more vehicle for expression of political sectarianism, making it another tool for the formation of public discourse and opinion (Neubaum and Krämer, 2017), political perceptions (Gunther, 1998; Lee and Tandoc, 2017), and political persuasion (Winter et al., 2015).

In light of the documented increase in distrust in media (Masullo et al., 2023), especially among conservatives (Levendusky et al., 2023) and the tendency of highly-partisan individuals to be more engaged with political expression (Bail, 2022; Mutz, 2006), we expect politicized reviews to be dominated by those perceiving some or all of the media to be biased against them and hostile to their views. Rojas (2010) suggested that individuals who perceive others as disproportionately affected by media, or view media as hostile to their side, are more likely to adopt new communication technologies to amplify their views. We therefore expect the politicization of news app reviews to be characterized by increased selectivity, hostility, and expression of political identity, all potential accelerators of polarization.

Selectivity, hostility, and media consumption as a political identity

Today's high-choice media environment offers many options for political and partisan consumers. Studies have shown that when given the choice, individuals tend to consume news that align with their preexisting political views (Mitchell et al., 2021; Stroud, 2008). This self-selection into congruent media can reinforce one's prior beliefs and strengthen a sense of belonging to a particular social identity (Young, 2019), which could intensify polarization and extremism (Mason, 2018). Scholars have been warning that selective exposure to like-minded information can create echo chambers and reduce tolerance for opposing viewpoints (Jamieson and Cappella, 2008), with more recent data demonstrating an asymmetry, where echo chambers, while rare on average, are more common on the American right (Levendusky et al., 2023). The relationship between polarization and selective exposure exhibits a reciprocal and reinforcing dynamic.

Partisans tend to selectively engage with media that mirror their political leanings, a process that in turn can amplify their endorsement of more extreme views (Iyengar et al., 2019; Stroud, 2010). Over time, this cyclic process can further heighten polarization (Rhodes, 2022).

In light of the above, our investigation into news app reviews on Google Play leads us to pose the first research questions:

RQ1. What are the predominant themes present in news app reviews on Google Play?

Given evidence for increased partisan polarization and emerging politicization of non-political online spaces, we hypothesize that:

H1. The prominence of politicized discourse in news app reviews on Google Play increased from 2009 to 2022.

With the rise in prominence of news media in general (Napoli, 2020) and partisan outlets specifically (Peck, 2019), mainstream news media institutions and journalists have faced growing scrutiny and criticism from politicians, pundits, and commentators (Ladd, 2011). The heightened attention to news channels and the positioning of journalists by conservatives as political actors (Carlson et al., 2021) have shaped public perceptions of journalistic professionalism and objectivity (Tsfati and Cappella, 2003). Studies have shown that common public perceptions, especially among conservatives, of a general liberal bias in news coverage were not a mere reflection of the content of news, but rather a response to frequent attacks from the political elites (Watts et al., 1999), and particularly conservatives and Republicans who for decades have practiced media-bashing as a political performance (Jamieson and Cappella, 2008). Former Republican President Donald Trump has further amplified these trends with relentless attacks and delegitimization campaigns that portrayed journalists using populist terms such as the enemy of the people and fake news media (Carlson et al., 2021). The increase in accusations of bias was accompanied by a dramatic increase in attacks on journalists both offline and online since 2015 (Wells et al., 2020).

Studies on the Hostile Media Bias (HMB) had consistently shown that perceptions of bias are subjective and can be influenced by political ideology (Hassell et al., 2020). Studies have demonstrated that partisans on both sides of arguments and ideological disputes tend to perceive neutral and balanced news articles as biased against their side (Vallone et al., 1985). The effect of HMB was particularly large when neutral articles were presented as being written by partisan outlets that were inconsistent with participants' political beliefs (Reid, 2012). As a result, while conservatives and Republicans are, on average, more likely to accuse the media of biases, such perceptions can also arise among liberals (e.g. Stroud, 2010). Recognizing that allegations of media bias, especially when examining multiple news apps across the political spectrum, can originate from both partisan sides, we ask the following research question:

RQ2. Does the prominence of politicized topics and themes vary across news app reviews for media outlets with different political leanings?

Specifically, given the effects of elites' rhetoric on public news trust and perceptions of media hostility (Van Duyn and Collier, 2019; Watts et al., 1999), we are particularly interested in content concerning bias criticism, hostility, and sentiment. Therefore, we ask:

RQ3. Does the political leaning of a news app correlated with the amount of bias criticism, hostility, and the sentiment used to write about them?

Finally, a key affordance of the app review section is the star rating system, which could potentially serve as a proxy for public opinion, and could, in turn, influence other users' consumption decisions (Kim et al., 2018). Audience feedback in general (Lee and Tandoc, 2017), and numeric ratings, in particular (Broersma, 2019), provide users with a proxy for the quality of products and public reaction to them (Lee and Tandoc, 2017). Since we believe politicization could be used to both endorse or criticize a news outlet (i.e. a very critical political comment is likely to be accompanied by a negative rating, but a review that praises the political view of the app would likely be accompanied by a positive score), we suggest a research question to examine the relationship between politicized language and numeric ratings of reviews, controlling for other coded factors, including hostility, bias criticism, and political leaning:

RQ4. How does the utilization of politicized language influence the numeric rating for news apps.

In addition, users have the option to upvote or downvote reviews, signaling their endorsement of or disagreement with a review (notice, the upvote is for the review itself, not the app). We expect politicization to appeal to some users but to be seen as a distraction or inappropriate in this context by others. We therefore pose the relationship between politicized language and upvoting as a research question and not a directional hypothesis:

RQ5. What is the relationship between politicization and the number of upvotes an app has received?

Methods

Data collection

Google Play reviews were scraped on 13 September 2022 for each news app posted between 2009 and 2022 using Python's Beautiful Soup module that parses web pages and provides a convenient interface for navigating content. The Google Play store was

Table 1. Summary on number of reviews across different political leanings on Google Play.

Political leaning	*N* of reviews Google Play	*N* of apps	Average *N* of each category (*SD*)	Range of reviews *N*
Right	140,460	14	10,033 (25,024)	10–96,015
Left	509,393	25	20,376 (32,845)	54–82,285
Moderate	109,290	7	15,613 (30,492)	2–140,980
Total	759,143	46	16,503(30,029)	2–140,980

officially launched in 2012, yet some of the apps were previously available via the Android store, and are still available for scraping, allowing us to go back to 2009 in our data (McIlroy et al., 2016).

We compiled a list of 46 news media outlets based on the criteria of appearing in both the Google Play store on the date of data collection, and in the political leaning classification provided by the AllSides media bias Chart, v7 from 2022 (https://www.allsides.com/media-bias/media-bias-chart). While no categorization into political slant is without flaws, AllSides was used in prior research for annotating the political leaning of news media (e.g. Matsumura et al., 2024; Rozado et al., 2022; Spinde et al., 2021). Nevertheless, mindful of the potential biases inherent in any categorization, we conducted a sensitivity analysis. This involves comparing the AllSides categorization with other sources, a process thoroughly detailed in Supplemental Appendix A. This comparison provided reassurance that our results are not significantly influenced by our choice of categorization method. Furthermore, while minor discrepancies in the degree of political bias (e.g. whether an outlet should be considered "leaning left" versus "left") were noted, there was a substantial agreement among various categorizations regarding the overall political leanings of the new outlets we studied. We therefore collapsed "leaning left" and "left" into "leaning left" and "leaning right" and "right" into "leaning right." This adjustment suits the theoretical framework of our study.

In total, we collected 759,143 reviews. As shown in Table 1, on average, left-leaning media outlets received the greatest number of reviews, followed by moderate, and then right media outlets. Our corpus consisted of 25 left-leaning, 7 centrist, and 14 right-leaning apps. Only very few disagreements were noted when comparing AllSides with other popular categorizations (see Supplemental Appendix A), providing us more confidence as to the stability of our analysis.

Measures

Thematic content. Our inductive analysis of topics and themes relied on the ANTMN framework (Walter and Ophir, 2019), which consists of four methodological steps: topic modeling, topic networking, community detection, and in-depth qualitative analysis. Topic modeling, in our case using Latent Dirichlet Allocation, is an unsupervised machine learning method used to identify themes in large textual data

(Blei, 2003). Topics are distribution lists of words and their probabilities based on co-occurrences of words in the same documents (here, Google Play reviews). The number of topics and the alpha hyperparameter (which accounts for the short length of the reviews) were chosen carefully based on a five-fold run over different combinations of κ (number of topics, between 5 and 100, in skips of 5) and α (0.01–0.5), after which we chose the best performing model using perplexity scores (see Ophir et al., 2023). While modeling in ANTMN is data-driven and is conducted inductively, the qualitative analysis of the top words and documents associated with each topic and their corresponding themes (clusters) is guided by the theoretical framework described earlier. The relative prominence of themes was also examined over time, providing a window into politicization dynamics. The optimal model, based on perplexity scores, had 35 topics with 2/35 alpha hyperparameter. For details of model selection, see Supplemental Appendix B.

In the second step of ANTMN, we calculated a network of topics based on their co-occurrence in the same reviews. This was done by computing the cosine similarity over the document topic matrix and using these similarities as links in the network. Each topic was treated as a node, and their co-occurrence formed the links. To reduce network density and discard non-significant links, we applied a backbone method (Serrano et al., 2009). In the third step, we used the Louvain community detection algorithm to group the topics into broader themes. Finally, we performed a qualitative analysis of the model and its output, and labeled each theme based on our close reading.

Manual coding. To gain additional insight into the nature of politicization, the three authors collaboratively developed a codebook based on the existing literature and then tested and refined it using small samples of reviews coded by two independent coders. The final analysis included sentiment of the review (positive vs negative), accusation of journalistic bias, and expressions of hostility toward journalists/media outlets. Three rounds of training, totaling 180 reviews, were conducted. The training samples were not used for the final analysis. The coders reached an adequate reliability for all variables in the last round of training (see Table 2). One exception was hostility, where we believe that the relatively lower reliability resulted from the rarity of the category in the corpus (i.e. a zero-inflated case). We thus recommend reading into the analysis of this specific variable with some caution. After reaching reliability, the two coders manually coded a random sample of 542 reviews.

Table 2. Intercoder reliability summary.

Variables	Percent agreement	Krippendorff's α	Cohen's κ	Scott's π
Inclusion	90	.80	.80	.80
Sentiment	90	.82	.82	.82
Biased criticism	100	1.00	1.00	1.00
Hostility toward journalists/media outlets	98.33	.66	.66	.66

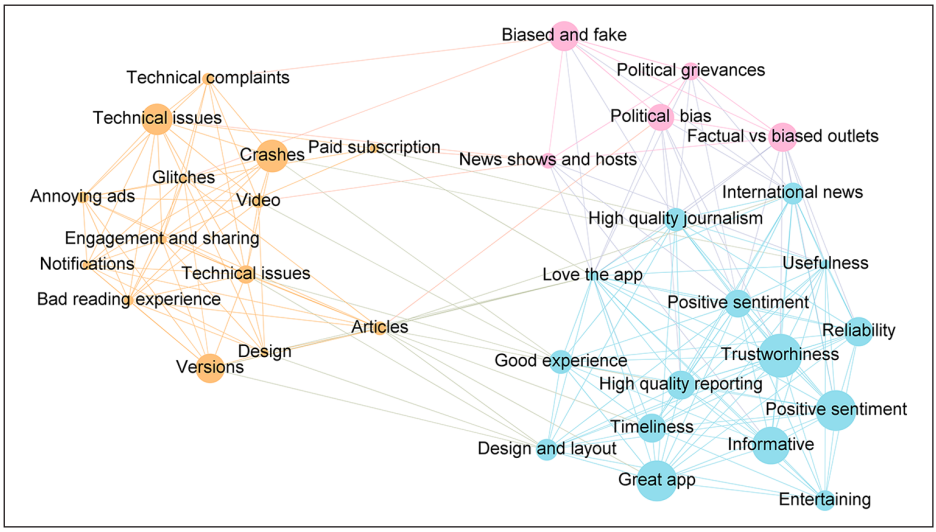

Figure 1. A topic model network where nodes are topics, and edges represent co-occurrence in posts. Node size indicates their prevalence and color represents their association with a theme based on Louvain community detection algorithm.

Results

Topics and themes

First, we analyzed the dominant themes in the corpus of Google Play reviews (RQ1). As shown in Figure 1, the model identified 35 topics that were clustered into three broad themes. Based on our in-depth qualitative analysis of each topic and their corresponding clusters, we labeled the themes Content (blue), Technical (orange), and Political (pink). The Political theme consisted of topics such as political bias in news coverage, aimed at other users and the journalists themselves, the Content theme focused on the quality of journalism, and the Technical theme focused on issues related to user experience, aimed at the development team. The top 10 unique words and texts per each topic can be found in the Supplemental Appendix C.

To gain additional confidence in our modeling and analyses, we further validated the inductive approach by comparing human and automated coding results, revealing that reviews marked as political by human coders contained more than double the politicized language as detected by the algorithm. Notably, non-political reviews still featured some political language, reflecting the inherent complexity of topic modeling. More information can be found in Supplemental Appendix D.

In line with the study's aims, our analysis primarily concerns the Political theme. We identified topics related to bias accusations, political discussions, and evaluation of the factuality of media outlets, shows, and hosts. For example, in the Factual versus Biased Outlets topic, reviewers often expressed the reasoning behind their selective exposure to specific outlets. For example, and in line with prior research showing asymmetry in

distrust in mainstream media, a conservative user argued that "the Epoch Times is one of the ONLY real news stations today! You can always count on them to give us the facts . . ." and another said "thank God for Fox News the only station that will report the actual news, not like the liberal media ABC NBC CBS MSNBC CNN News only Fox News is trusted."

As expected, some topics were dedicated to explicitly blame incongruent outlets of being biased or fake, such as the biased and fake, political bias, and political grievances topics. Reviews on these topics often referred to specific political events or politicians. For instance, on the biased and fake topic, a review attacked CNN, stating that

> where President Trump is concerned, CNN makes biased, false, and half-true reports, in order to further its own agenda of delegitimizing him, and to create, based upon these false and inflammatory reports, anger and fear in the minds of our citizens. This is not news, it's propaganda of the worst kind.

Similarly, another review criticized CNN's coverage of the 2020 election, saying, "swamp media, I am a Republican but this network is fake news just like the others and I am sure owned and controlled by the same people Terrible news. How they cover the 2020 election and post-election is criminal."

Users who used the political bias topic often compared their favorite channel to what they perceived as inferior others. One post highlighted that compared with AP and NPR, Fox News catered to narrow sub-populations only, saying that

> the information Fox uses for their content is selectively chosen to appeal to the conservative right-leaning audience. They don't discuss things objectively and let inherent bias and emotions devalue their content. This is not a good news source for people of this country, while sources like AP news and NPR offer much better objective journalism and reporting.

Another review accused Yahoo! News of distorting political facts toward a liberal perspective: "Yahoo has joined hands with other liberal media and instead of just reporting political facts, distort the facts to promote a liberal agenda. Where can we get just the facts anymore?!"

Users relying on the political grievances topic often discussed political affairs unrelated to the media that cover them. One review for OAN made false claims about the integrity of the 2020 elections and used derogatory language to refer to President Biden, stating that people should continue to

> refer to Donald J Trump as pres. If u believe Alzbiden [sic. This is a common derogatory and false reference to Biden allegedly suffering from Alzheimer's] got more votes than Obama in 1st election and Reagan in 84, you should be put in the same care home Biden is obviously well on his way too, because you are obviously delusional or you live in a little fantasy world. On another note why is it when the real press was in office c19 death were over 2Mil but now that Joe Alzhiden the Walter look alike is in we magically have only 600k never seen him n Walter n same room coincidence. No.

Another review explicitly called for violence, reasoning that

> I honestly am worried about the direction this country is going. I realize that abortion is killing babies, but I still think it is right for some nonsensical reason. Also, did anybody want trump? No! I realize that Hillary sucked but she should've been elected. My fellow liberals! Band together with me as we take up arms against the filthy conservatives. #impeach trump then rip off his hair.

Furthermore, some users criticized outlets like AP for perceived bias of their reporting, accusing them of withholding certain stories or presenting information with a slant,

> It's not what stories AP reports it's what they don't. On Trump for example they don't report or investigate all of the good stories. And when they do its slanted to make a positive a negative. Showing their slant. I don't trust anything on AP since I learned the board that approves what stories are reported had changed. IT'S [sic] was once 2 democrats, 2 republicans & 1 libertarian, to represent the population. Since Bill Clinton's first term it has been made up of five democrats. AND THEY NEVER TOLD US. Shame on those who knew and said nothing. There are two sides to any story but just one set of facts. The slant comes from the side of the store the reporter believes before covering the facts.

Frequency and change over time

Over time, there has been a consistent increase in use of the Content and Political theme at the expense of the technical one from 2009 to 2022, supporting H1. Importantly, at the beginning of our dataset, political discussions are almost completely absent. But a decade later, politics constituted over 20% of the content, almost as much as technical issues, an area of discussion that by this point has dramatically plummeted. In addition to a linear upward trend, focus on the Political theme also corresponded to real-world political events in the United States and the world, as depicted in Figure 2 (lower panel). While we see a slow increase in the early 2010s, politicization began to more sharply

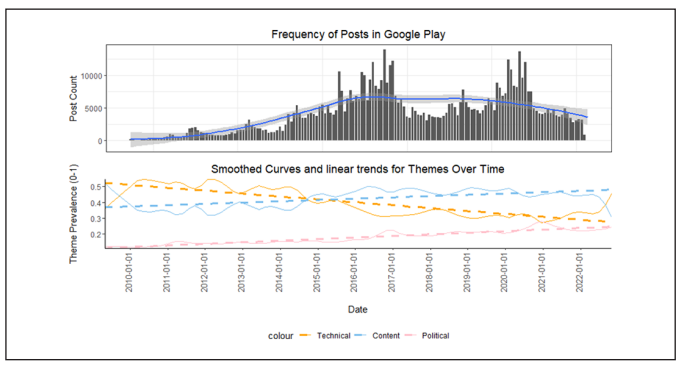

Figure 2. News reviews themes over time, including frequency (upper panel) and smoothed thematic prominence (lower panel).
To clarify, the figure does not depict the fluctuations in the quantity of politicized and non-politicized reviews over time. Rather, it shows the relative share of politicized language in reviews over time, independent of the total number of reviews.

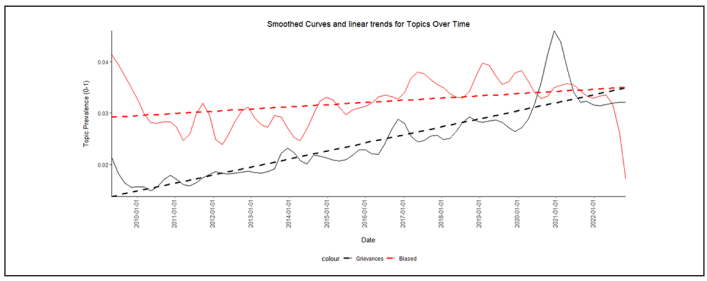

Figure 3. Topical prevalence over time of *biased and fake* and *political grievances* topics.

grow in mid-2015, around the time Donald Trump announced his presidential candidacy. Two substantial peaks can also be seen around the 2016 and 2020 elections.

To gain additional insight into the news politicization, we analyzed topics marked by hostility toward media: biased and fake and political grievances. These topics, centered around allegations of political bias, provided insights into shifts in users' perceptions toward news content. As shown in Figure 3, both topics demonstrated upward trends over time. The political grievances topic notably surged around the January 6th Capitol riot, when supporters of Donald Trump violently invaded the Capitol building following Donald Trump's refusal to accept his defeat in the 2020 presidential election, which was based on misinformation and conspiracy theories (Eggers et al., 2021). We postulate that these unsubstantiated claims for voter fraud and the violent events that followed likely contributed to the increased prevalence of politically charged reviews in this period.

In terms of frequency, we observe spikes in activity around specific dates, which can largely be explained by 2016 and 2020 US presidential elections. One of the most significant spikes in activity was recorded in July 2016 when Donald Trump secured the necessary delegates to officially become the Republican Party's nominee for the US presidency. Another notable spike was in October 2020, during which key events such as the first presidential debate, President Trump's COVID-19 response, and the confirmation hearings for Supreme Court nominee Amy Coney Barrett.

Politicization across different media outlets categories

RQ2 examined whether reviews for media outlets of different political slant exhibited different levels of politicization. An analysis of variance of manual-coding reviews ($n=542$) was conducted to examine the relationship between political slant and share of politicized language in reviews. The results showed a significant relationship between the political leaning of media outlets and political posts, $F(2,238)=5.931$, $p=.003$. Further analysis using Tukey's *post hoc* test indicated that right-leaning media received the highest share of politicized language in reviews ($M=0.458$, $SD=0.256$) followed by left-leaning media ($M=0.340$, $SD=0.277$, $p=.001$), and centrist media ($M=0.289$, $SD=0.237$, $p=.000$). The difference between centrist and right-leaning media was significant ($p=.006$), as was the difference between right- and left-leaning media ($p=.011$). The difference in politicization between left-leaning and centrist media was not significant ($p=.537$).

Table 3. Regression results of media outlets' political leaning, Political theme, hostility toward media and journalist, and biased criticism on star ratings.

	Coefficient (*SE*)	Standardized coefficient
Intercept	4.750 (0.199)***	
Left-leaning news outlets	−0.059 (0.206)	−.017
Conservative news outlets	−0.173 (0.237)	−.0046
Biased criticism	−2.244 (0.227)***	−.547
Hostility toward journalists and media	−1.013 (0.295)***	−.189
Political theme	−0.481 (0.304)	.080
N	235	
Adjusted R^2	.53	

***$p < .001$.

Sentiment, perceived bias, and media hostility

RQ3 probed differences in criticism, hostility, and sentiment among reviews for right, left, and centrist news apps. We relied on the manual coding to investigate the sentiment, perceived bias and media hostility. Ignoring posts centered on technical aspects or unrelated to politics, our sample of 542 reviews revealed 45% contained political content. Of these, 172 had positive sentiments, 66 were negative, 8 neutral, 21.54% accused apps of bias, and 10.57% showed media hostility. Probit regressions indicated that there were no significant differences across political alignments (all $p > .05$).

RQ4 asked whether the star-system rating (1–5) was affected by the political leanings of media outlets, prevalence of the Political theme, expressions of hostility toward media and journalist, and accusations of bias (sentiment was not used as a predictor to avoid a tautology). Regression results in Table 3 revealed that using right-leaning news outlets as the comparison category, left-leaning ($p = .96$) and centrist news outlets ($p = .96$) did not have significant impact on star ratings. However, biased criticism ($p < .00$) and media hostility ($p = .00$) negatively impacted star ratings, with bias criticism playing a major role. There was no significant impact of Political theme on star ratings ($p = .116$).

Finally, RQ5 asked the relationship between political reviews and the number of upvotes. Given that most reviews had few upvotes (Median = 0, $M = 2.424$, Max = 2745), we used a zero-inflated Poisson regression to analyze the data. Results indicated that more political reviews were less likely to get upvotes from other users ($b = -0.722$).

Discussion

Driven by identity politics, Americans have increasingly aligned themselves with two homogeneous groups—conservative Republicans or liberal Democrats (Mason, 2018). This identity sorting has resulted in spillovers of polarization and politicization into non-political (at least in the partisan sense) spaces, from health and science to interpersonal relationships and entertainment. In this study, we offer a novel approach to studying politicization of non-political spaces. Relying on a mixed-method approach that combining inductive unsupervised machine learning methods and deductive, theory-driven

qualitative analysis (Walter and Ophir, 2019), we investigated whether the user-generated news app reviews on Google Play, which is intended for technical and user-experience discussions, has become more politicized and polarized from 2009 to 2022.

Our findings provide alarming evidence for a rapidly growing politicization of the ostensibly non-political space of reviews in Google Play. We identified 35 topics that were automatically clustered into three themes using the ANTMN approach (Walter and Ophir, 2019), which we qualitatively labeled Political, Content, and Technical. While the Technical theme consisted of topics in which users approach developers with complaints and suggestions have to do with the user experience of the app itself (as intended by Google Play's rules), the Political theme was loaded with discussions of journalistic quality, and particularly political bias. The Content theme was preoccupied with the quality of journalism, but was not necessarily about politics, and rarely included direct attacks on the integrity of media.

The Content and Political themes were often used in the same reviews, while the technical discussions were usually separated. Nevertheless, our method was able to isolate the Political theme, enabling us to focus our analysis on this explicitly politicized language cluster. These analyses, in turn, demonstrated that over the years, the share of reviews' language associated with politics increased dramatically. Some scholars have argued that in the absence of clear institutional slants, non-political online spaces could be expected to provide an environment for citizens to engage in everyday political talk, and facilitate more diverse and deliberative conversations (Wojcieszak and Mutz, 2009; Yan et al., 2018). However, our findings revealed the existence of politicized reviews that tended to focus on political bias and political grievances. These results suggest that within the context of journalism, political discussions in non-political spaces may also veer toward polarization, as individual identities can be triggered by the media brands themselves as part of broader culture wars.

Various factors may have contributed to this trend. Distrust in mainstream media (asymmetrically higher among conservatives, see Benkler et al., 2018), was often fueled by the attacks from political figures, especially conservatives and Republicans (Jamieson and Cappella, 2008), and by media polarization (Klar and Krupnikov, 2016; Levendusky and Malhotra, 2016). Studies have demonstrated that the growing polarization in America has been driven more by affect than by substantial changes in policy support, expressed via a growing dislike and intolerance of the opposite party (Krupnikov and Ryan, 2022). We expected the politicization of non-political spaces (e.g. Ophir et al., 2023) to make its way to news apps reviews as well, based on prior research showing news consumers who actively follow politics are also more likely to express their political opinions (Iyengar et al., 2019) and strong partisans being more vocal online than moderates (Bail, 2022).

Interestingly, in addition to the linear trend toward politicization, we also observed temporary peaks in Political theme during significant political event, such as presidential elections. These spikes could possibly be explained by the agenda-setting theory (McCombs and Shaw, 1972), and in the case of reviews could perhaps be thought of as examples of a unique type of inter-media agenda setting processes (Harder et al., 2017). Furthermore, exposure to politicized reviews might influence public opinion and perceptions (Gunther, 1998), which could in turn potentially amplify perceived polarization.

We found politicization to be specifically high when users reviewed non-centrist outlets. Given identity-driven selective exposure (Peterson et al., 2021) and media outlets' inclination to pick sides (Ladd, 2011), it is unsurprising to see partisan outlets receiving more politicized reviews than centrist media, which at least claim to represent both sides. It should be noted that we did not directly observe selective exposure, or exposure in general. One may leave a review on CNN's app decrying them to be "fake news" without ever using the app or reading any of its journalistic content. We do, however, speculate that the politicization of app reviews may lead to selective news consumption, which could, potentially, bolster polarization. Future studies may look more closely at the relationship between reviews and actual consumption using complementary methods.

In our analysis, we did not find significant differences in hostility, sentiment, or bias accusations across various political news outlets. Content factors, however, impacted app ratings, which were found in prior studies to be key drivers of users' behavior (Chen, 2017; Margolin and Markowitz, 2018). Our results showed that while the political leanings of media outlets did not sway star ratings in our data, one prominent Political theme did negatively correlate with them. Users often discussed politics to convey criticism or negative sentiments. Notably, reviews highlighting bias or hostility were associated with lower ratings. As our data indicate a rise over time in accusations of bias, these could have long-term effects on news consumption and polarization. Given the bandwagon heuristic, where majority opinions influence perceptions of quality (Metzger and Flanagin, 2013), overall star ratings could override individual preferences (Messing and Westwood, 2014) and further undermine the trust in news media and the political system.

Interestingly, users in our corpus seem reluctant, on average, to encounter political content in app reviews. According to Google's guidelines, the reviews page is intended for users to communicate with app developers, and the company explicitly advises against including politics in reviews. This policy was a primary reason for us to choose the platform for this study. Our data suggest users found politicized reviews less valuable, given the lesser upvotes they received, on average. Though we cannot infer intentions or the psychological mechanisms underlying the observed behaviors, this finding may suggest that users preferred app reviews to be non-politicized, as intended by Google. Future studies may use complementary methods, such as surveys or interviews, to better understand users' reaction to politicization.

Some limitations of this study are acknowledged and provide opportunities for future research. First, we focused exclusively on Google Play reviews due to data availability and legal constraints in accessing large datasets from platforms like Apple's App Store (Sadiq et al., 2021). Even though our research doesn't encompass all app stores, Google Play's vast database of over a decade offers a significant starting point. During the years studied, the Google Play Store had the largest database compared with other app stores. Most importantly, the Google Play store made an explicitly call to avoid political discourse in reviews, thus making it the most appropriate choice for our research questions into the politicization of non-political spaces. Another advantage is that the Google Play store has a relatively lax approach to content moderation, compared with Apple's store (Schmidt-Kraepelin et al., 2019). Future research can expand the dataset to collect news app reviews from diverse app stores that covering a broader user base. Future studies

may compare our findings with other stores, countries, and contexts. Second, as mentioned above, we didn't explore the motivations behind political discussions in app reviews. Addressing such questions via triangulation of additional methods, such as surveys, interviews, and focus groups, could shed light on political discourse in unexpected online spaces. Future studies may thus add valuable insight into the motivations and outcomes of political discussions in seemingly non-political online spaces. Finally, our study is the first to examine the politicization of news apps, and as such, we focused on the American context only. However, politicization and polarization are global phenomena (e.g. Fischer, 2022), and we encourage subsequent studies to adapt our methodologies for the study of other countries and contexts, enriching the comparative understanding of these issues.

In sum, this study investigated the growing politicization of a non-political space, specifically the news app review section on Google Play, using a mixed-method approach. Like in other areas of contemporary public life, our findings point to alarming and steadily growing levels of politicization and polarization. We call for scholars, journalists, and practitioners to pay more attention to consumers' linguistic behavior, expressed not only in comments on social media or news websites, but in unpredictable spaces such as the reviews section on app stores, to better understand the audience feedback. More work into the effects of such politicization would allow for a better understanding of the growing hostility toward media and journalism in the United States and around the world, a key threat to democratic societies in the 21st century. Like other work into user-generated content online, our findings suggest that a reexamination of questions around content moderation may be in order as well. In line with the lack of empirical research, minimal attention was given by regulators and scholars to the politicization, and potentially hate speech and incitement of violence, in non-political spaces such as the app review section. We hope that our study will encourage the direction of more attention and resources to such problems in the future.

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ORCID iD

Rui Wang https://orcid.org/0000-0001-6535-1694

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Note

1. https://www.pewresearch.org/short-reads/2022/03/10/the-polarization-in-todays-congress-has-roots-that-go-back-decades/

References

Bail C (2022) *Breaking the Social Media Prism: How to Make Our Platforms Less Polarizing*. Princeton, NJ: Princeton University Press.

Bakardjieva M (2015) Do clouds have politics? Collective actors in social media land. *Information, Communication & Society* 18(8): 983–990.

Bedingfield S (2018) "Who is Nicholas Stanford": the New York Times music critic and his secret role in the rise of the "liberal media" claim. *American Journalism* 35(4): 398–419.

Benkler Y, Faris R and Roberts H (2018) *Networked Propaganda: Manipulation, Disinformation, and Radicalization in American Politics*. Oxford: Oxford University Press.

Blei DM (2003) Latent Dirichlet allocation. *Journal of Machine Learning Research* 3: 993–1022.

Broersma M (2019) Audience engagement. In: Vos TP, Hanusch F, Dimitrakopoulou D, et al. (eds) *The International Encyclopedia of Journalism Studies*. 1st ed. Hoboken, NJ: Wiley, pp. 1–6.

Carlson M, Robinson S and Lewis SC (2021) *News After Trump: Journalism's Crisis of Relevance in a Changed Media Culture*. Oxford: Oxford University Press.

Chen A and McCabe KT (2022) Roses and thorns: political talk in reality TV subreddits. *New Media & Society*.

Chen C-W (2017) Five-star or thumbs-up? The influence of rating system types on users' perceptions of information quality, cognitive effort, enjoyment and continuance intention. *Internet Research* 27(3): 478–494.

Eggers AC, Garro H and Grimmer J (2021) No evidence for systematic voter fraud: a guide to statistical claims about the 2020 election. *Proceedings of the National Academy of Sciences* 118(45): e2103619118.

Finkel EJ, Bail CA, Cikara M, et al. (2020) Political sectarianism in America. *Science* 370(6516): 533–536.

Fischer S (2022) Distrust in political, media and business leaders sweeps the globe. *Axios*, 18 January. Available at: https://www.axios.com/2022/01/18/distrust-in-political-media-and-business-leaders-sweeps-the-globe

Google Play (2023). Available at: https://play.google.com/about/comment-posting-policy/?hl=en-US

Gunther AC (1998) The persuasive press inference: effects of mass media on perceived public opinion. *Communication Research* 25(5): 486–504.

Hanisch C (1969) The personal is political. Available at: https://www.carolhanisch.org/CHwritings/PIP.html

Harder RA, Sevenans J and Van Aelst P (2017) Intermedia agenda setting in the social media age: how traditional players dominate the news agenda in election times. *The International Journal of Press/Politics* 22(3): 275–293.

Hassell HJG, Holbein JB and Miles MR (2020) There is no liberal media bias in which news stories political journalists choose to cover. *Science Advances* 6(14): eaay9344.

Huber GA and Malhotra N (2017) Political homophily in social relationships: evidence from online dating behavior. *The Journal of Politics* 79(1): 269–283.

Iyengar S and Westwood SJ (2015) Fear and loathing across party lines: new evidence on group polarization. *American Journal of Political Science* 59(3): 690–707.

Iyengar S, Lelkes Y, Levendusky M, et al. (2019) The origins and consequences of affective polarization in the United States. *Annual Review of Political Science* 22(1): 129–146.

Jamieson KH and Cappella JN (2008) *Echo Chamber: Rush Limbaugh and the Conservative Media Establishment*. Oxford: Oxford University Press.

Keefe J (2020) Quiz: can you tell a "Trump" fridge from a "Biden" fridge? *The New York Times*, 28 October. Available at: https://www.nytimes.com/interactive/2020/10/27/upshot/biden-trump-poll-quiz.html

Kim JM, Jun M and Kim CK (2018) The effects of culture on consumers' consumption and generation of online reviews. *Journal of Interactive Marketing* 43: 134–150.

Klar S and Krupnikov Y (2016) *Independent Politics*. Cambridge: Cambridge University Press.

Kor-Sins R (2023) The alt-right digital migration: a heterogeneous engineering approach to social media platform branding. *New Media & Society* 25: 2321–2338.

Krupnikov Y and Ryan JB (2022) *The Other Divide: Polarization and Disengagement in American Politics*. Cambridge: Cambridge University Press.

Ladd JM (2011) *Why Americans Hate the Media and How It Matters*. Princeton, NJ: Princeton University Press.

Lee A (2020) Trader Joe's democrats and Walmart republicans. *Medium*, 29 September. Available at: https://towardsdatascience.com/are-you-a-trader-joes-democrat-or-a-walmart-republican-a7b156131435

Lee E-J and Tandoc EC (2017) When news meets the audience: how audience feedback online affects news production and consumption: when news meets the audience. *Human Communication Research* 43(4): 436–449.

Levendusky M (2013) *How Partisan Media Polarize America*. Chicago, IL: University of Chicago Press. Available at: https://press.uchicago.edu/ucp/books/book/chicago/H/bo16468853.html

Levendusky M and Malhotra N (2016) Does media coverage of partisan polarization affect political attitudes? *Political Communication* 33(2): 283–301.

Levendusky M, Pasek J, Holbert RL, et al. (2023) *Democracy Amid Crises: Polarization, Pandemic, Protests, and Persuasion*. Oxford: Oxford University Press.

Margolin D and Markowitz DM (2018) A multitheoretical approach to big text data: comparing expressive and rhetorical logics in yelp reviews. *Communication Research* 45(5): 688–718.

Mason L (2018) *Uncivil Agreement: How Politics Became Our Identity*. Chicago, IL: University of Chicago Press.

Masullo GM, Tenenboim O and Lu S (2023) "Toxic atmosphere effect": uncivil online comments cue negative audience perceptions of news outlet credibility. *Journalism* 24: 101–119.

Matsumura A, Garg R, Hussain M, et al. (2024) Political orientation of online media sources and reporting of Covid-19 vaccine myocarditis. *PLoS ONE* 19(1): e0296295.

McCombs ME and Shaw DL (1972) The agenda-setting function of mass media. *Public Opinion Quarterly* 36(2): 176–187.

McIlroy S, Ali N and Hassan AE (2016) Fresh apps: an empirical study of frequently-updated mobile apps in the Google play store. *Empirical Software Engineering* 21(3): 1346–1370.

Messing S and Westwood SJ (2014) Selective exposure in the age of social media: endorsements Trump partisan source affiliation when selecting news online. *Communication Research* 41(8): 1042–1063.

Metzger MJ and Flanagin AJ (2013) Credibility and trust of information in online environments: the use of cognitive heuristics. *Journal of Pragmatics* 59: 210–220.

Mitchell A, Jurkowitz M, Oliphant B, et al. (2021) About a quarter of republicans, democrats consistently turned only to news outlets whose audiences aligned with them politically in 2020. *Pew Research Center's*, 22 February. Available at: https://www.pewresearch.org/journalism/2021/02/22/about-a-quarter-of-republicans-democrats-consistently-turned-only-to-news-outlets-whose-audiences-aligned-with-them-politically-in-2020/

Mouffe C (2005) *On the Political*. London: Psychology Press.

Munson S and Resnick P (2011) The prevalence of political discourse in non-political blogs. *Proceedings of the International AAAI Conference on Web and Social Media* 5(1): 1.

Mutz DC (2001) The future of political communication research: reflections on the occasion of Steve Chaffee's retirement from Stanford University. *Political Communication* 18(2): 231–236.

Mutz DC (2006) *Hearing the Other Side: Deliberative Versus Participatory Democracy*. Cambridge: Cambridge University Press.

Napoli L (2020) *Up All Night: Ted Turner, CNN, and the Birth of 24-Hour News*. New York: Harry N. Abrams.

Neubaum G and Krämer NC (2017) Opinion climates in social media: blending mass and interpersonal communication: opinion climates in social media. *Human Communication Research* 43(4): 464–476.

Ophir Y, Pruden ML, Walter D, et al. (2023) Weaponizing reproductive rights: a mixed-method analysis of white nationalists' discussion of abortions online. *Information, Communication & Society* 26(11): 2186–2211.

Peck R (2019) *Fox Populism: Branding Conservatism as Working Class.* Cambridge University Press.

Peterson E, Goel S and Iyengar S (2021) Partisan selective exposure in online news consumption: evidence from the 2016 presidential campaign. *Political Science Research and Methods* 9(2): 242–258.

Radwell S and Israel J (2021) *American Schism: How the Two Enlightenments Hold the Secret to Healing our Nation*. Austin, TX: Greenleaf Book Group.

Rajadesingan A, Budak C and Resnick P (2021) Political discussion is abundant in non-political subreddits (and less toxic). *Proceedings of the International AAAI Conference on Web and Social Media* 15: 525–536.

Reid SA (2012) A self-categorization explanation for the hostile media effect. *Journal of Communication* 62(3): 381–399.

Rhodes SC (2022) Filter bubbles, echo chambers, and fake news: how social media conditions individuals to be less critical of political misinformation. *Political Communication* 39(1): 1–22.

Riedl MJ, Lukito J and Woolley SC (2023) Political influencers on social media: an introduction. *Social Media + Society* 9(2): 20563051231177938.

Rojas H (2010) "Corrective" actions in the public sphere: how perceptions of media and media effects shape political behaviors. *International Journal of Public Opinion Research* 22(3): 343–363.

Rozado D, Hughes R and Halberstadt J (2022) Longitudinal analysis of sentiment and emotion in news media headlines using automated labelling with transformer language models. *PLoS ONE* 17(10): e0276367.

Sadiq S, Umer M, Ullah S, et al. (2021) Discrepancy detection between actual user reviews and numeric ratings of Google App store using deep learning. *Expert Systems with Applications* 181: 115111.

Schmidt-Kraepelin M, Thiebes S and Sunyaev A (2019) Investigating the relationship between user ratings and gamification–a review of mHealth apps in the apple app store and Google Play store. Available at: https://scholarspace.manoa.hawaii.edu/server/api/core/bitstreams/893dfeeb-3e93-4e5e-ab62-cbfbb5bf4c13/content

Serrano MÁ, Boguñá M and Vespignani A (2009) Extracting the multiscale backbone of complex weighted networks. *Proceedings of the National Academy of Sciences* 106(16): 6483–6488.

Shin J, Yang A, Liu W, et al. (2022) Mask-wearing as a partisan issue: social identity and communication of party norms on social media among political elites. *Social Media + Society* 8(1): 20563051221086233.

Spinde T, Rudnitckaia L, Mitrović J, et al. (2021) Automated identification of bias inducing words in news articles using linguistic and context-oriented features. *Information Processing & Management* 58(3): 102505.

Stroud NJ (2008) Media use and political predispositions: revisiting the concept of selective exposure. *Political Behavior* 30(3): 341–366.

Stroud NJ (2010) Polarization and partisan selective exposure. *Journal of Communication* 60(3): 556–576.

Suk J, Zhang Y, Yue Z, et al. (2023) When the personal becomes political: unpacking the dynamics of sexual violence and gender justice discourses across four social media platforms. *Communication Research* 50(5): 610–632.

Sunstein CR (2018) *#Republic: Divided Democracy in the Age of Social Media*. Princeton, NJ: Princeton University Press.

Tsfati Y and Cappella JN (2003) Do people watch what they do not trust? Exploring the association between news media skepticism and exposure. *Communication Research* 30(5): 504–529.

Turner JC (2010) Social categorization and the self-concept: a social cognitive theory of group behavior. In: Postmes T and Branscombe NR (eds) *Rediscovering social identity*. London: Psychology Press, pp. 243–272.

Vallone RP, Ross L and Lepper MR (1985) The hostile media phenomenon: biased perception and perceptions of media bias in coverage of the Beirut massacre. *Journal of Personality and Social Psychology* 49(3): 577.

Van Boven L, Ehret PJ and Sherman DK (2018) Psychological barriers to bipartisan public support for climate policy. *Perspectives on Psychological Science* 13(4): 492–507.

Van Duyn E and Collier J (2019) Priming and fake news: the effects of elite discourse on evaluations of news media. *Mass Communication and Society* 22(1): 29–48.

Walter D and Ophir Y (2019) News frame analysis: an inductive mixed-method computational approach. *Communication Methods and Measures* 13(4): 248–266.

Walter D, Ophir Y and Jamieson KH (2020) Russian Twitter accounts and the partisan polarization of vaccine discourse, 2015–2017. *American Journal of Public Health* 110(5): 718–724.

Walter D, Ophir Y, Lokmanoglu AD, et al. (2022) Vaccine discourse in white nationalist online communication: a mixed-methods computational approach. *Social Science & Medicine* 298: 114859.

Watts MD, Domke D, Shah DV, et al. (1999) Elite cues and media bias in presidential campaigns: explaining public perceptions of a liberal press. *Communication Research* 26(2): 144–175.

Wells C, Shah D, Lukito J, et al. (2020) Trump, Twitter, and news media responsiveness: a media systems approach. *New Media & Society* 22(4): 659–682.

Winter S, Brückner C and Krämer NC (2015) They came, they liked, they commented: social influence on Facebook news channels. *Cyberpsychology, Behavior, and Social Networking* 18(8): 431–436.

Wojcieszak ME and Mutz DC (2009) Online groups and political discourse: do online discussion spaces facilitate exposure to political disagreement? *Journal of Communication* 59(1): 40–56.

Yan W, Sivakumar G and Xenos MA (2018) It's not cricket: examining political discussion in nonpolitical online space. *Information, Communication & Society* 21(11): 1571–1587.

Young DG (2019) *Irony and Outrage: The Polarized Landscape of Rage, Fear, and Laughter in the United States*. Oxford: Oxford University Press.

Author biographies

Rui Wang is a PhD candidate in the Department of Communication at the University at Buffalo. She studies communication technology, political communication, and misinformation. Combining

computational methods and quantitative methods, her work focuses on the examination of politicization and polarization within online public discourses. She also examines the psychological factors of public perceptions toward the application of artificial intelligence (AI).

Sagarika Suresh Thimmanayakanapalya is a PhD candidate at the University at Buffalo in the field of information systems with a quantitative machine learning research orientation. She is keen on understanding the design of and building embodied social agents that lie at the intersection of generative AI and Metaverse. She currently works on encapsulating safe measurement sciences for rapidly evolving AI digital landscapes through consumer research and developing low-cost fidelity information systems artifacts.

Yotam Ophir PhD is an Assistant Professor of Communication at the University at Buffalo. His work combines computational methods for text mining, network analysis, experiments, and surveys to study media content and effects in the areas of political, science, and health communication. He is the head of the Media Effects, Misinformation, and Extremism lab and a member of the Center for Information Integrity at the University at Buffalo.